

CHEMISTRY 101 · WEEK 2 QUIZ

Chemistry 101 — Week 2

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. In chemistry, what makes something an acid?

- ☐ A. It contains extra oxygen atoms, which is what makes it fizz.
- ☐ B. It lets go of loose hydrogen pieces, which is what tastes sour.
- ☐ C. It has been heated until its molecules move around faster.
- ☐ D. It is made mostly of carbon atoms, like everything living.

2. On the pH scale, which number is neutral — neither acid nor base?

- ☐ A. 0
- ☐ B. 14
- ☐ C. 10
- ☐ D. 7

3. Vinegar is about pH 3 and milk is about pH 6. Roughly how many times more acidic is the vinegar?

- ☐ A. About 3 times, one for each step down the scale.
- ☐ B. About 100 times, because pH steps double each time.
- ☐ C. About 1,000 times, because each pH step is ten times.
- ☐ D. About 30 times, ten for the first step and ten more.

4. What is the normal pH range of blood?

- ☐ A. 5.50 to 7.50
- ☐ B. 7.00 to 8.00
- ☐ C. 7.35 to 7.45
- ☐ D. 6.35 to 6.45

5. What does carbon dioxide turn into when it dissolves in blood?

- ☐ A. Pure oxygen, which the hemoglobin carries.
- ☐ B. Carbonic acid, which lowers the blood's pH.
- ☐ C. Bicarbonate, which raises the blood's pH.
- ☐ D. Lactic acid, which the muscles then burn.

6. A preemie's blood gas shows a pH of 7.20 and a CO_2 that is much too high. What is the most likely problem, and the fix?

- ☐ A. The lungs aren't clearing CO_2 – give more breathing help.
- ☐ B. The cells are starving for oxygen – start antibiotics.
- ☐ C. The baby is breathing too much – turn the ventilator down.
- ☐ D. The kidneys are losing bicarbonate – give more fluids.

7. What is bicarbonate's job in the blood?

- ☐ A. It turns sugar into energy inside the mitochondria.
- ☐ B. It soaks up loose hydrogen so acid can't do damage.
- ☐ C. It makes the blood thicker so it clots when cut.
- ☐ D. It carries oxygen from the lungs to every cell.

8. Where was the pH scale invented, and by whom?

- ☐ A. At a brewery in Denmark, by the chemist Søren Sørensen (1909).
- ☐ B. At a hospital in France, by the doctor Louis Pasteur (1860).
- ☐ C. At a university in England, by the chemist Joseph Priestley (1774).
- ☐ D. At a soda factory in America, by the chemist Charles Alderton (1885).

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)